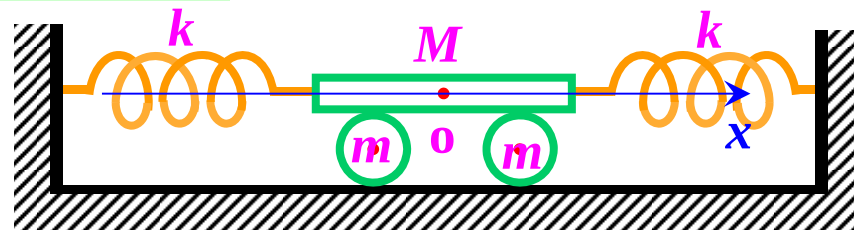


二、用能量法求解简谐振动的角频率

方法三 圆筒纯滚动, 摩擦力均不做功, 系统机械能守恒



$$2\frac{1}{2}kx^2 + \frac{1}{2}Mv_M^2 + 2\left(\frac{1}{2}mv_{mC}^2 + \frac{1}{2}J_C\Omega^2\right) = E \quad J_C = mr^2$$

圆筒纯滚动

$$v_{mC} = \Omega r$$

$$v_M = v_{mC} + \Omega r = 2v_{mC}$$

$$\Omega = \frac{v_{mC}}{r} = \frac{v_M}{2r}$$

$$kx^2 + \frac{1}{2}Mv_M^2 + 2\left(\frac{1}{2}m\frac{1}{4}v_M^2 + \frac{1}{2}mr^2\Omega^2\right) = E$$

$$kx^2 + \frac{1}{2}Mv_M^2 + 2\left(\frac{1}{2}m\frac{1}{4}v_M^2 + \frac{1}{2}m\frac{1}{4}v_M^2\right) = E$$

$$kx^2 + \frac{1}{2}Mv_M^2 + \frac{1}{2}mv_M^2 = E$$

$$kx^2 + \frac{1}{2}(M + m)v_M^2 = E$$

机械能守恒, 不随时间而变 $\frac{dE}{dt} = 0$ 例5.4有疑问 1

$$kx^2 + \frac{1}{2}(M+m)v_M^2 = E \quad 2kx \frac{dx}{dt} + \frac{1}{2}(M+m)2v_M \frac{dv_M}{dt} = 0$$

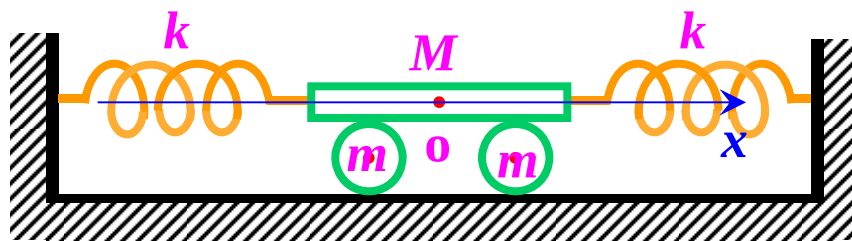
$$2kxv_M + \frac{1}{2}(M+m)2v_M a_M = 0$$

ω^2

$$2kx + (M+m)a_M = 0 \quad a_M = -\frac{2k}{M+m}x \quad \text{简谐振动}$$

$$\omega^2 = \frac{2k}{M+m} \quad \omega = \sqrt{\frac{2k}{M+m}}$$

$$T = 2\pi \sqrt{\frac{M+m}{2k}}$$



机械能守恒不能作为简谐振动的判据!

讨论

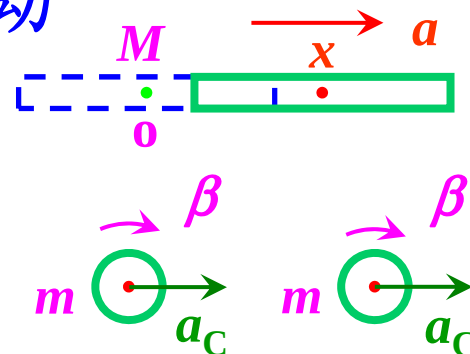
圆筒的运动 (1) 圆筒质心的运动

由伽利略变换, 平板加速度

$$a = \beta r + a_C = 2\beta r = 2a_C$$

平板速度 $v = \Omega r + v_C = 2v_C$

平板位移 $x = r\theta + x_C = 2x_C$



$$a = -\frac{2k}{M+m}x$$

$$a = 2a_C = -\frac{2k}{M+m}x = -\frac{2k}{M+m}2x_C$$

$$a_C = -\frac{2k}{M+m}x_C$$

ω^2

圆筒质心的运动是同频率的简谐振动

(2) 圆筒纯滚动

$$a_C = \beta r$$

$$x_C = r\theta$$

$$\beta = \frac{a_C}{r} = -\frac{2k}{(M+m)}x_C \cdot \frac{1}{r} = -\frac{2k}{(M+m)}\theta$$

ω^2

圆筒的转动也是同频率的简谐振动



§ 5.4 阻尼振动

§ 5.5 受迫振动 共振

自学!

§ 5.6 振动的合成

一、同方向同频率简谐振动的合成

设物体沿 x 轴同时
参与两个独立振动:

$$x_1 = A_1 \cos(\omega t + \varphi_1)$$

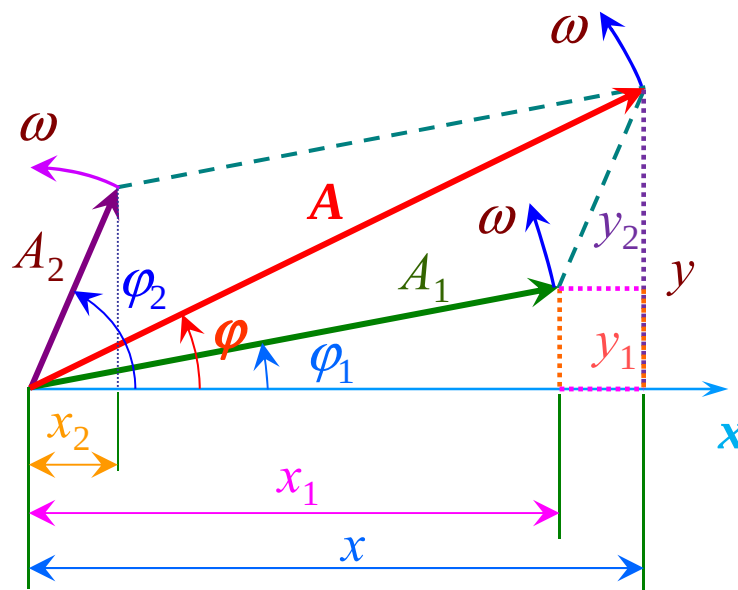
$$x_2 = A_2 \cos(\omega t + \varphi_2)$$

则质点的合位移为:

$$x = x_1 + x_2 = A_1 \cos(\omega t + \varphi_1) + A_2 \cos(\omega t + \varphi_2)$$

由旋转矢量可知合成后仍然为简谐振动, 设合振动的振动表达式为:

$$x = A \cos(\omega t + \varphi)$$



其中合振动的振幅和初相可从矢量图求出：

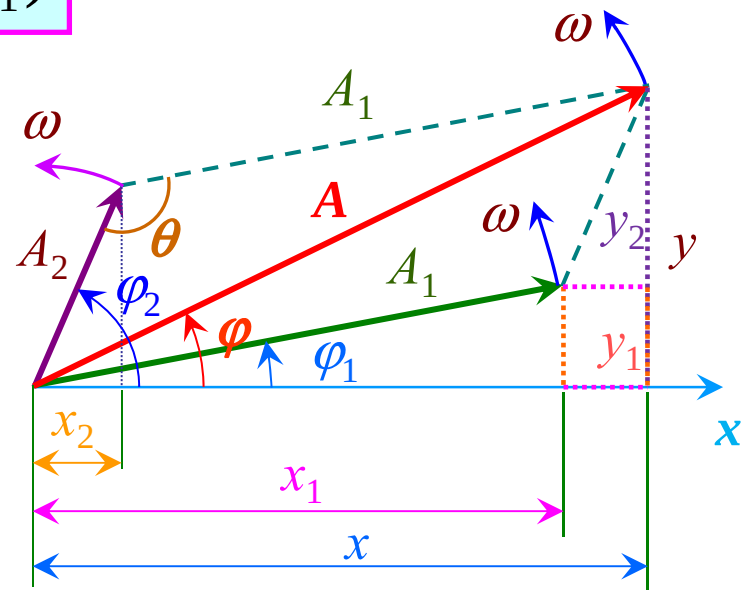
$$A = \sqrt{A_1^2 + A_2^2 - 2A_1A_2 \cos \theta}$$
$$= \sqrt{A_1^2 + A_2^2 - 2A_1A_2 \cos[\pi - (\varphi_2 - \varphi_1)]}$$

$$A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos(\varphi_2 - \varphi_1)}$$

$$\tan \varphi = \frac{y}{x} = \frac{y_1 + y_2}{x_1 + x_2}$$

$$\tan \varphi = \frac{A_1 \sin \varphi_1 + A_2 \sin \varphi_2}{A_1 \cos \varphi_1 + A_2 \cos \varphi_2}$$

两个值

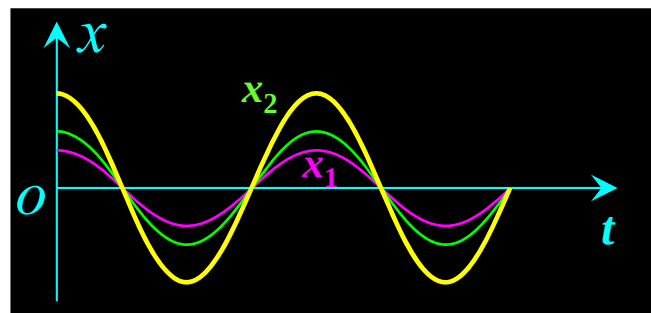
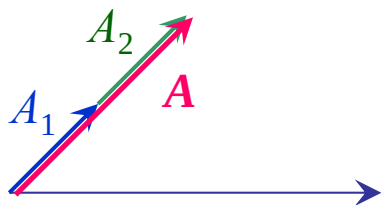


讨论

$$A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos(\varphi_2 - \varphi_1)}$$

① x_1 和 x_2 同相, 即

当 $\varphi_2 - \varphi_1 = 2m\pi$, ($m = 0, \pm 1, \pm 2, \dots$) 时,
$$\begin{cases} A = A_1 + A_2 \\ \varphi = \varphi_2 \text{ or } \varphi = \varphi_1 \end{cases}$$

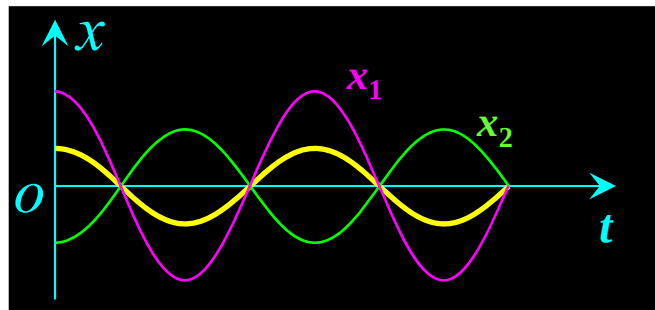
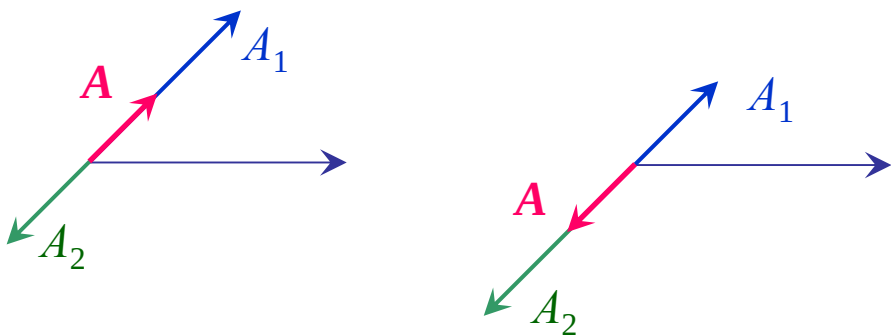


特别当 $A_1 = A_2$ 时, 合振幅为分振幅的两倍, 能量为4倍。

② x_1 和 x_2 反相, 即

当 $\varphi_2 - \varphi_1 = (2m+1)\pi$, ($m = 0, \pm 1, \pm 2, \dots$) 时,

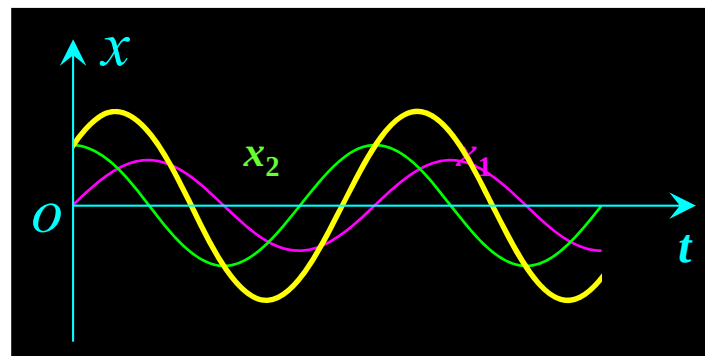
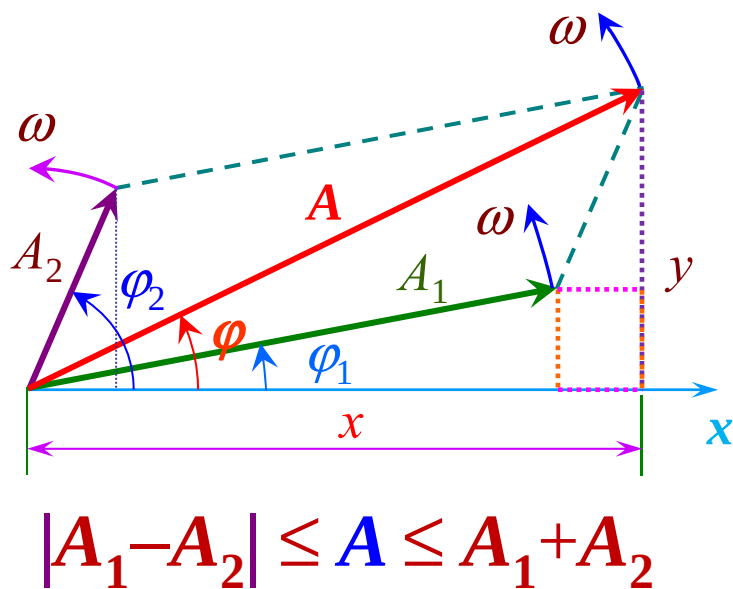
合振动振幅: $A = |A_1 - A_2|$



合振动初相位: 如果 $A_1 > A_2$, $\varphi = \varphi_1$; 若 $A_1 < A_2$, $\varphi = \varphi_2$

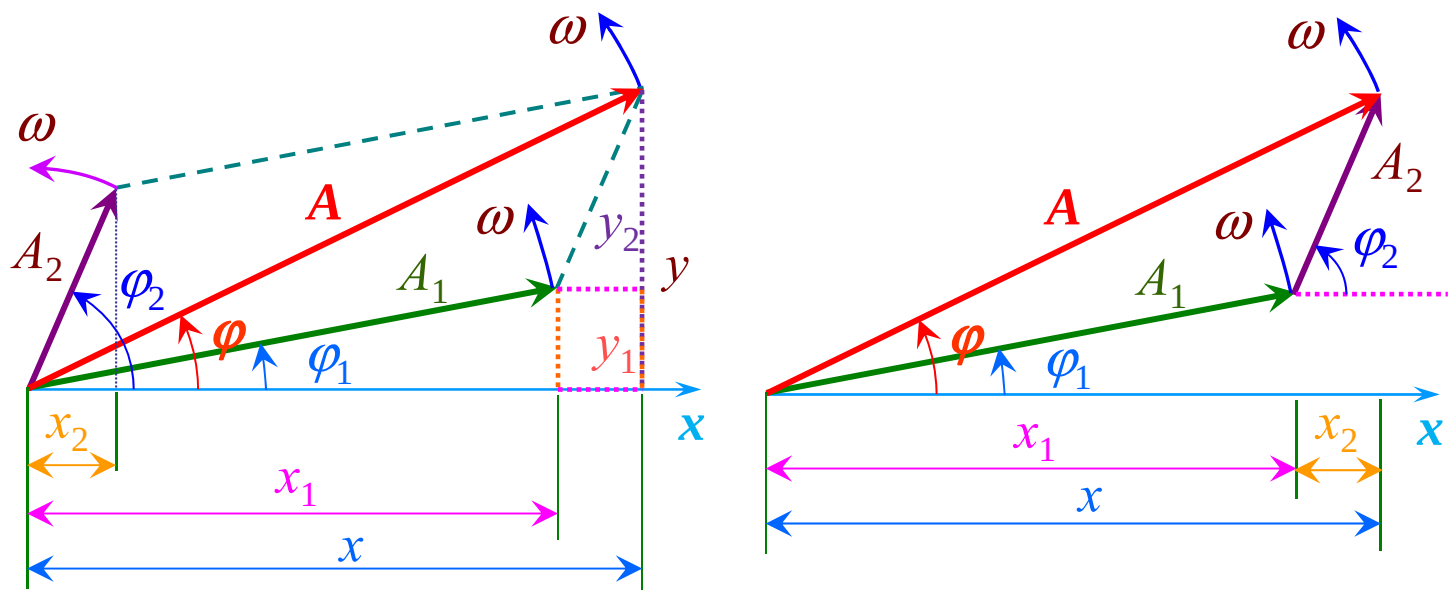
特别当 $A_1 = A_2$ 时, 合振幅为零, 能量也最小.

③ x_1 和 x_2 相位差为任意值



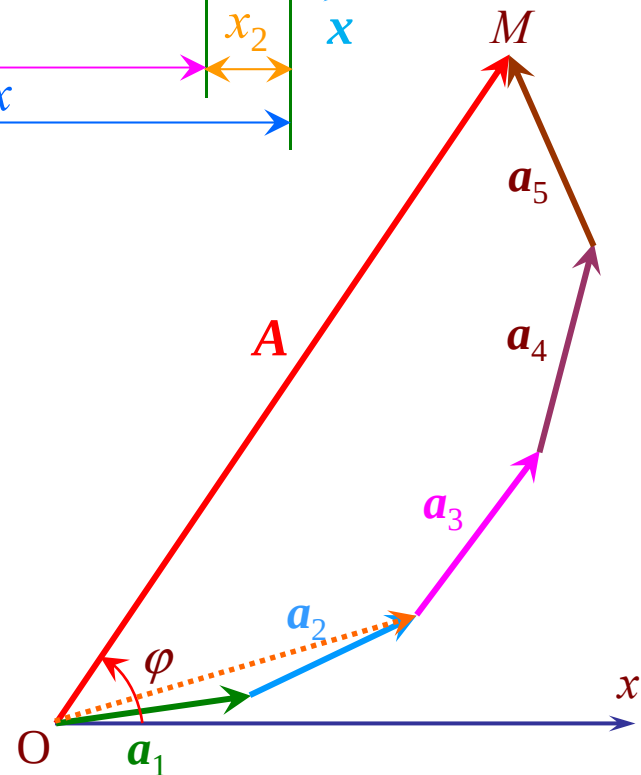
合振动的初相可借助旋转矢量图求出.

④ 振动合成旋转矢量图的变形:

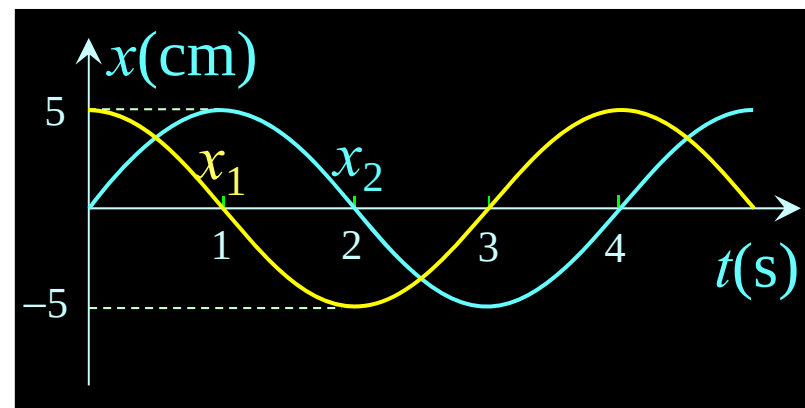


⑤ 对于多个简谐振动的合成, 同样可用旋转矢量法迭加而得:

注意振动矢量和其旋转矢量表示图之间的差别



例：两个频率和振幅都相同的简谐振动，其 $x-t$ 曲线如图所示，求：(1)两个简谐振动的相位差；
(2)两个简谐振动合成后的振动方程。

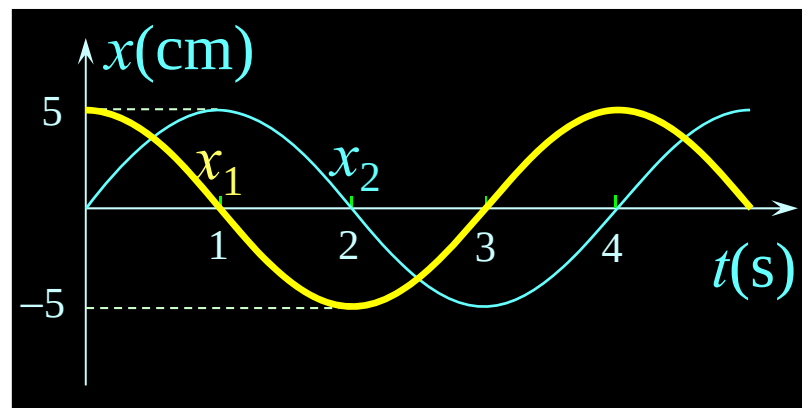


例：两个频率和振幅都相同的简谐振动，其 $x-t$ 曲线如图所示，求：(1)两个简谐振动的相位差；
(2)两个简谐振动合成后的振动方程。

解：由图可得

$$A_1 = A_2 = 5 \text{ cm} \quad T_1 = T_2 = T = 4 \text{ s}$$

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{4} = \frac{\pi}{2} \text{ rad} \cdot \text{s}^{-1}$$



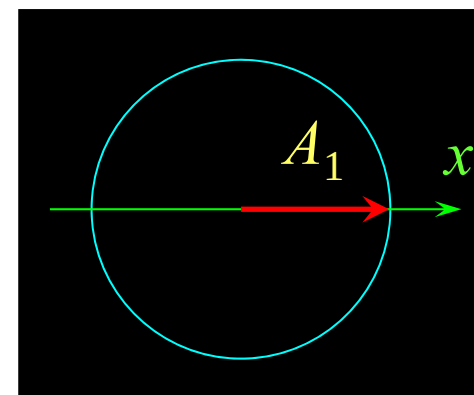
(1) 两个简谐振动的相位差

设 $x_1 = A_1 \cos(\omega t + \varphi_1) = 5 \cos(\omega t + \varphi_1)$

对于 x_1 , $t=0$ 时, $x_{10} = 5 \text{ cm} > 0$

$$5 = 5 \cos(\omega \times 0 + \varphi_1) \quad \cos \varphi_1 = 1$$

$$\varphi_1 = 0 \quad x_1 = 0.05 \cos\left(\frac{\pi}{2} t\right) (\text{SI})$$



设 $x_2 = A_2 \cos(\omega t + \varphi_2) = 5 \cos(\omega t + \varphi_2)$

对于 x_2 , $t=0$ 时, $x_{20}=0$ cm

$$0 = 5 \cos(\omega \times 0 + \varphi_2)$$

$$\cos \varphi_2 = 0$$

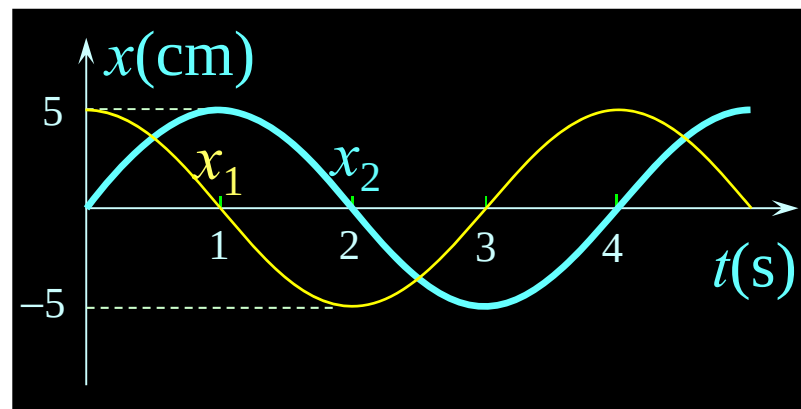
$$\varphi_2 = \frac{\pi}{2} \quad \text{或} \quad \varphi_2 = -\frac{\pi}{2}$$

对于 x_2 , $t=0$ 时, $v_{20} > 0$

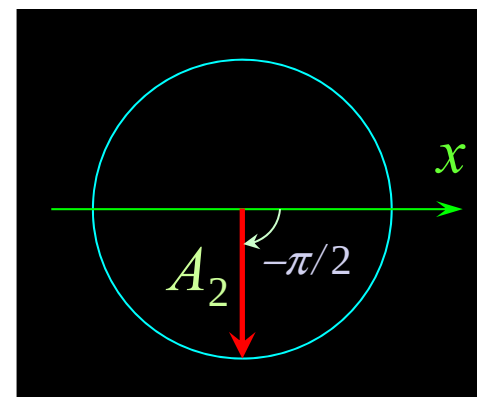
$$x_2 = 0.05 \cos\left(\frac{\pi}{2}t - \frac{\pi}{2}\right) \text{ (SI)}$$

两个简谐振动的相位差

$$\Delta\varphi = \omega t + \varphi_2 - (\omega t + \varphi_1) = -\frac{\pi}{2} - 0 = -\frac{\pi}{2}$$



$$\varphi_2 = -\frac{\pi}{2}$$



(2) 两个简谐振动合成后的振动方程

$$A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \Delta\varphi}$$

$$= \sqrt{5^2 + 5^2 + 2 \times 5 \times 5 \times \cos(-\frac{\pi}{2})} = 5\sqrt{2} \text{ cm}$$

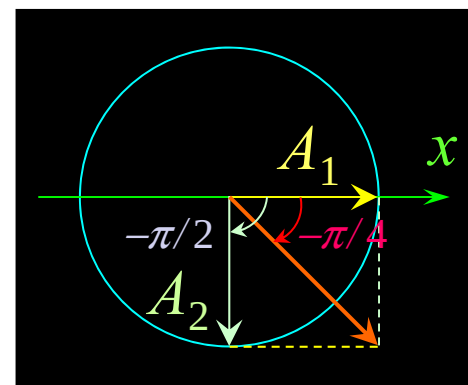
$$\tan \varphi = \frac{A_1 \sin \varphi_1 + A_2 \sin \varphi_2}{A_1 \cos \varphi_1 + A_2 \cos \varphi_2} = \frac{5 \times 0 + 5 \times (-1)}{5 \times 1 + 5 \times 0} = -1$$

$$\varphi = -\frac{\pi}{4} \quad \text{或} \quad \varphi = \frac{3\pi}{4}$$

两个简谐振动合成后 $-\frac{\pi}{2} < \varphi < 0$

$$\varphi = -\frac{\pi}{4}$$

$$x = 0.05\sqrt{2} \cos(\frac{\pi}{2}t - \frac{\pi}{4}) \text{ (SI)}$$



二. 同方向不同频率两简谐振动的合成 拍

物体沿x轴同时 $x_1 = A_1 \cos(\omega_1 t + \varphi_1)$
参与两个独立振动, $x_2 = A_2 \cos(\omega_2 t + \varphi_2)$

相位差: $\Delta\phi = (\omega_2 t + \varphi_2) - (\omega_1 t + \varphi_1)$

合振动不是简谐振动, 合振动为:

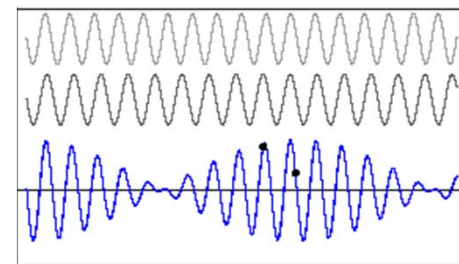
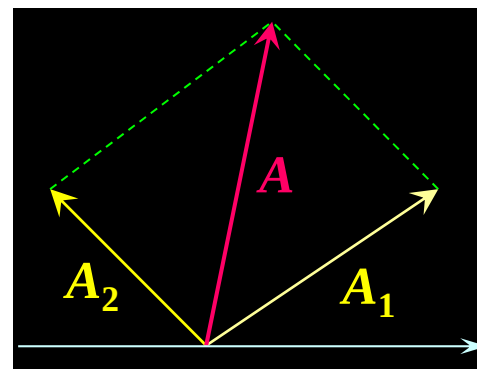
$$x = x_1 + x_2 = A_1 \cos(\omega_1 t + \varphi_1) + A_2 \cos(\omega_2 t + \varphi_2)$$

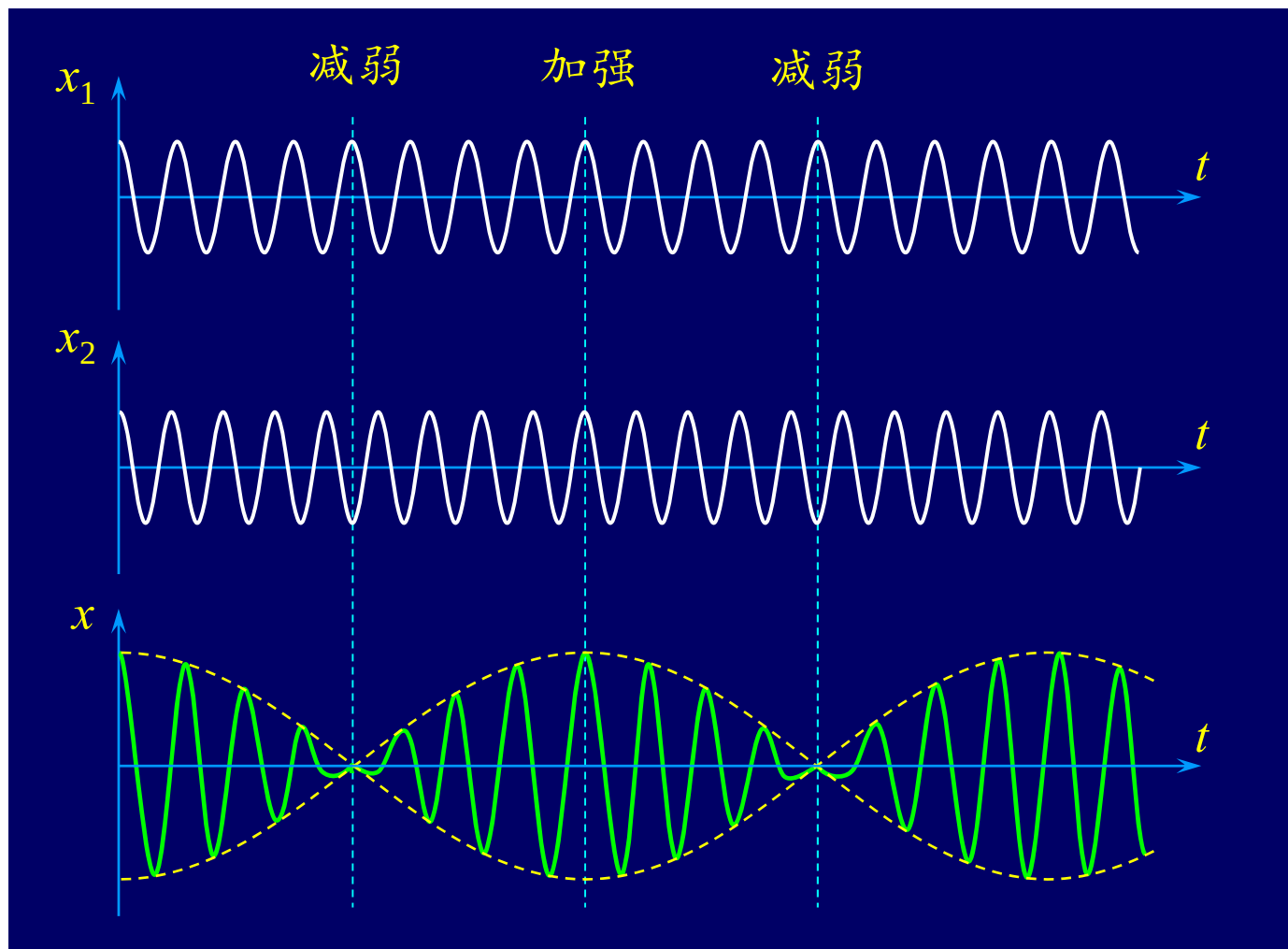
为简化分析过程: 设 $A_1 = A_2 = a$, $\varphi_1 = \varphi_2 = 0$

则:
$$x = 2a \cos\left[\left(\frac{\omega_2 - \omega_1}{2}\right)t\right] \cos\left[\left(\frac{\omega_2 + \omega_1}{2}\right)t\right]$$

上式中第一项
$$A = 2a \left| \cos\left[\left(\frac{\omega_2 - \omega_1}{2}\right)t\right] \right|$$

可看作随时间变化的振幅.





乐器校音

$$A = 2a \left| \cos \left[\left(\frac{\omega_2 - \omega_1}{2} \right) t \right] \right|$$

上式表明, 两个频率相近的简谐振动合成时, 合振动的振幅已不再保持恒定, 而是做缓慢的周期性改变, 合振幅的这种周期性的变化称为“**拍**”. 单位时间内振幅强弱的变化次数称为“**拍频**”相当于:

$$\cos \left[\left(\frac{\omega_2 - \omega_1}{2} \right) t \right] = \pm 1 \quad \frac{|\omega_2 - \omega_1|}{2} \Delta t = \frac{|\omega_2 - \omega_1|}{2} T_{\text{拍}} = \pi$$

$$T_{\text{拍}} = \frac{2\pi}{|\omega_2 - \omega_1|} \quad \text{拍频为:}$$

$$\nu_{\text{拍}} = \frac{1}{T_{\text{拍}}} = \left| \frac{\omega_2}{2\pi} - \frac{\omega_1}{2\pi} \right| = |\nu_2 - \nu_1|$$

三、相互垂直的简谐振动的合成

1、两相互垂直, 同频率简谐振动的合成

设质点同时参与分别在 x 、 y 方向上进行的两个振动, 它们的振动表达式分别为:

$$\begin{cases} x = A_1 \cos(\omega t + \varphi_1) \\ y = A_2 \cos(\omega t + \varphi_2) \end{cases}$$

则质点的运动方程为:

$$\vec{r} = x \vec{i} + y \vec{j} = A_1 \cos(\omega t + \varphi_1) \vec{i} + A_2 \cos(\omega t + \varphi_2) \vec{j}$$

此合成振动的振幅应限制在 $2A_1$ 和 $2A_2$ 的矩形内, 在直角坐标系中的轨道方程因为:

$$\frac{x^2}{A_1^2} + \frac{y^2}{A_2^2} - 2 \frac{xy}{A_1 A_2} \cos(\varphi_1 - \varphi_2) = \sin^2(\varphi_1 - \varphi_2)$$

证明: $\frac{x}{A_1} = \cos \omega t \cos \varphi_1 - \sin \omega t \sin \varphi_1$ ①

$$\frac{y}{A_2} = \cos \omega t \cos \varphi_2 - \sin \omega t \sin \varphi_2 \quad ②$$

以 $\cos \varphi_2$ 和 $\cos \varphi_1$ 分别乘①②两式,然后相减,得:

$$\frac{x}{A_1} \cos \varphi_2 - \frac{y}{A_2} \cos \varphi_1 = \sin \omega t \sin(\varphi_2 - \varphi_1) \quad ③$$

以 $\sin \varphi_2$ 和 $\sin \varphi_1$ 分别乘①②两式,然后相减,得:

$$\frac{x}{A_1} \sin \varphi_2 - \frac{y}{A_2} \sin \varphi_1 = \cos \omega t \sin(\varphi_2 - \varphi_1) \quad ④$$

③④两式
平方后
相加,得:

$$\frac{x^2}{A_1^2} + \frac{y^2}{A_2^2} - 2 \frac{xy}{A_1 A_2} \cos(\varphi_1 - \varphi_2) = \sin^2(\varphi_1 - \varphi_2)$$

证
毕

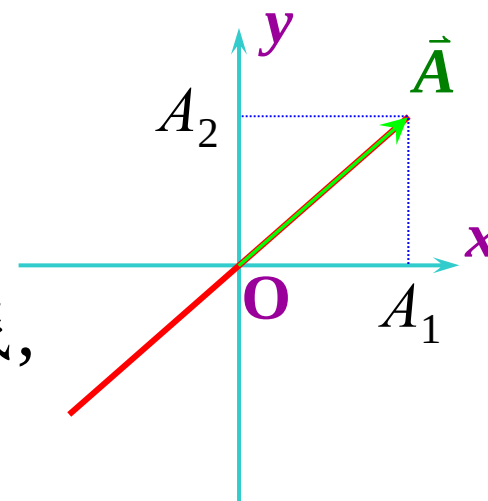
$$\frac{x^2}{A_1^2} + \frac{y^2}{A_2^2} - 2 \frac{xy}{A_1 A_2} \cos(\varphi_1 - \varphi_2) = \sin^2(\varphi_1 - \varphi_2)$$

讨论:

① a) $\varphi_2 - \varphi_1 = 0$ 时 (或 $2k\pi$), 上述轨道方程变为:

$$\left(\frac{x}{A_1} - \frac{y}{A_2} \right)^2 = 0 \quad y = \frac{A_2}{A_1} x$$

这是一条通过原点的I、III象限直线,
质点离开平衡位置的位移满足方程:



$$\vec{r} = x \vec{i} + y \vec{j}$$

$$= A_1 \cos(\omega t + \varphi_1) \vec{i} + A_2 \cos(\omega t + \varphi_1) \vec{j}$$

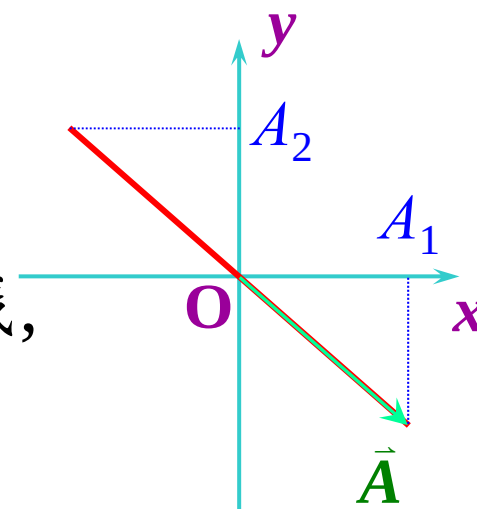
$$\vec{r} = (A_1 \vec{i} + A_2 \vec{j}) \cos(\omega t + \varphi_1)$$

$$\frac{x^2}{A_1^2} + \frac{y^2}{A_2^2} - 2\frac{xy}{A_1 A_2} \cos(\varphi_1 - \varphi_2) = \sin^2(\varphi_1 - \varphi_2)$$

b) $\varphi_2 - \varphi_1 = \pi$ 时[或 $(2k+1)\pi$], 上述轨道方程变为:

$$\left(\frac{x}{A_1} + \frac{y}{A_2} \right)^2 = 0 \quad y = -\frac{A_2}{A_1} x$$

这是一条通过原点的II、IV象限直线,
质点离开平衡位置的位移满足方程:



$$\vec{r} = x \vec{i} + y \vec{j}$$

$$= A_1 \cos(\omega t + \varphi_1) \vec{i} + A_2 \cos(\omega t + \varphi_1 + \pi) \vec{j}$$

$$\vec{r} = (A_1 \vec{i} - A_2 \vec{j}) \cos(\omega t + \varphi_1)$$

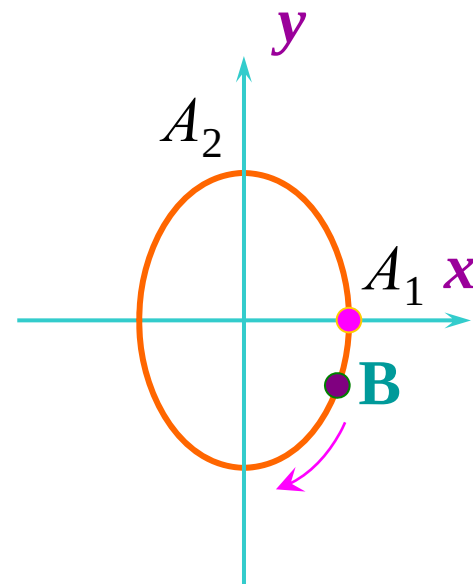
$$\frac{x^2}{A_1^2} + \frac{y^2}{A_2^2} - 2\frac{xy}{A_1A_2}\cos(\varphi_1 - \varphi_2) = \sin^2(\varphi_1 - \varphi_2)$$

② $\varphi_2 - \varphi_1 = \frac{\pi}{2}, \frac{3\pi}{2}$ 或 $(-\frac{\pi}{2})$, 或 $(2k+1)\frac{\pi}{2}$ 时, **轨道为椭圆**

$$\frac{x^2}{A_1^2} + \frac{y^2}{A_2^2} = 1 \quad \text{不是简谐振动}$$

当 $A_1 = A_2$ 、且 $\varphi_2 - \varphi_1 = \pm\frac{\pi}{2}$ 时,.

$$x^2 + y^2 = A_1^2 = A_2^2 \quad \text{为圆轨道}$$



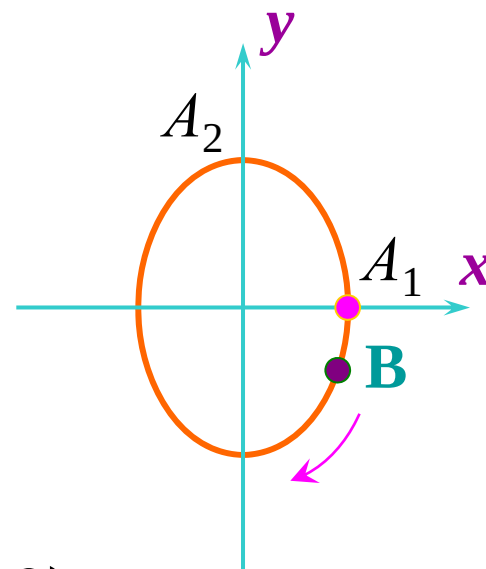
椭圆或圆振动时, **$\pi > \varphi_2 - \varphi_1 > 0$** 对应顺时针运动
 质点的运动方向 **$-\pi < \varphi_2 - \varphi_1 < 0$** 对应逆时针运动

质点运动方向的确定:

(a) $\varphi_2 - \varphi_1 = +\pi/2$ 时为**顺时针**方向转动.

$$\text{设 } x = A_1 \cos(\omega t + \varphi)$$

$$y = A_2 \cos(\omega t + \varphi + \pi/2)$$



假设 $t = t_0$ 时刻, $\omega t_0 + \varphi = 0$, 质点 $(A_1, 0)$

则 $t = t_0 + \Delta t$ 时(设 Δt 较小), $\pi/2 > \omega t + \varphi > 0$, $x > 0$,

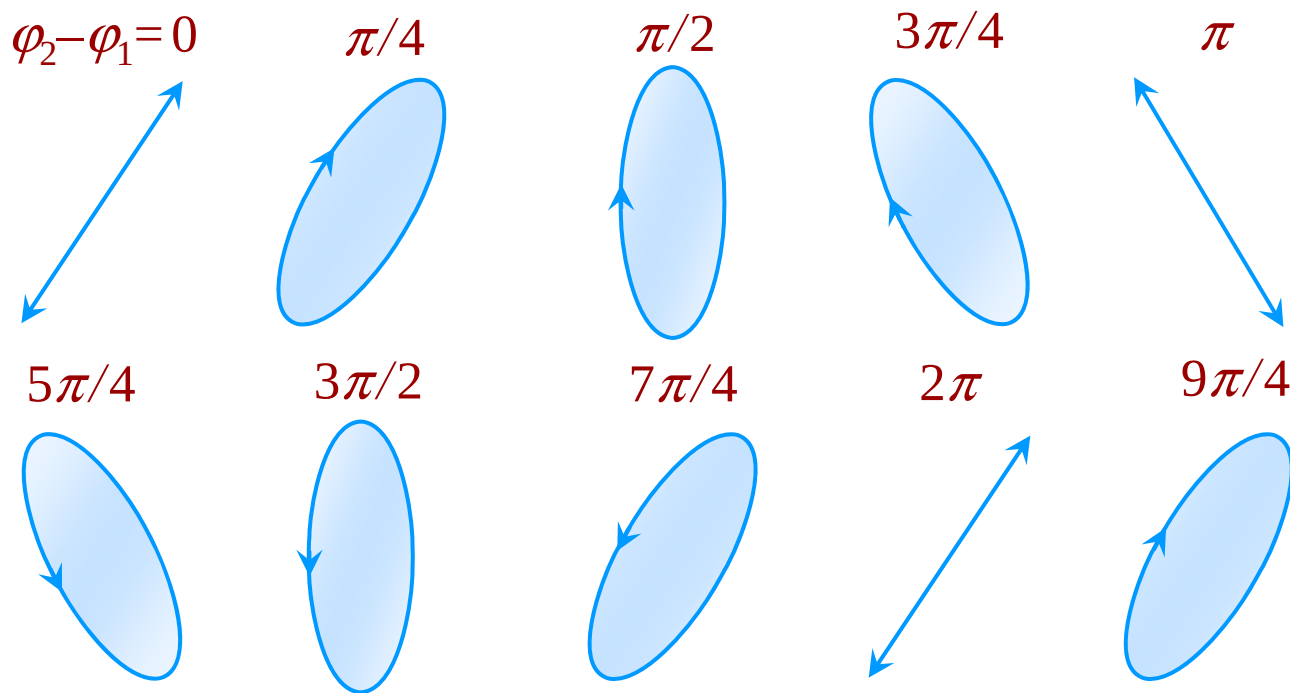
$$\pi > \omega t + \varphi + \pi/2 > \pi/2, y < 0.$$

质点由 $(A_1, 0) \rightarrow B$, **顺时针**方向转动

(b) $\varphi_2 - \varphi_1 = -\pi/2$ 时为**逆时针**方向转动.

$$\frac{x^2}{A_1^2} + \frac{y^2}{A_2^2} - 2\frac{xy}{A_1A_2}\cos(\varphi_1 - \varphi_2) = \sin^2(\varphi_1 - \varphi_2)$$

③ 当两个分振动的相位差 $\varphi_2 - \varphi_1$ 为任意值时, 物体作一般的椭圆运动



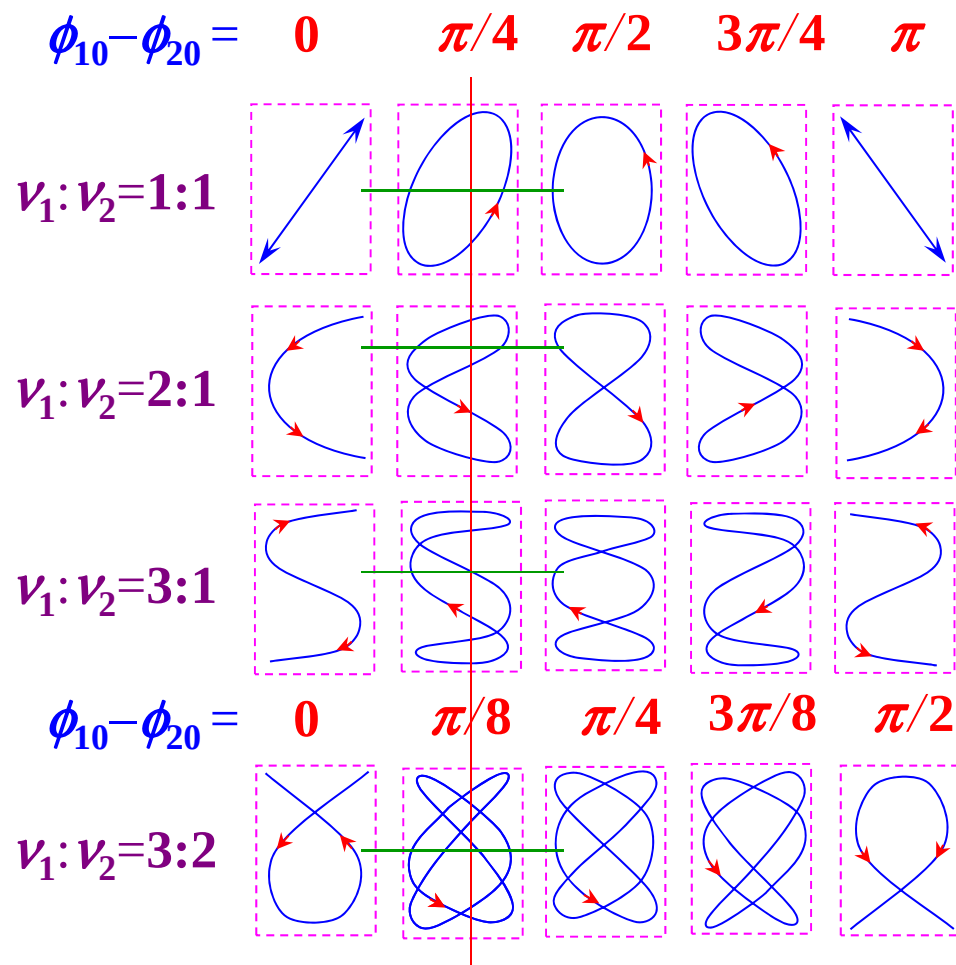
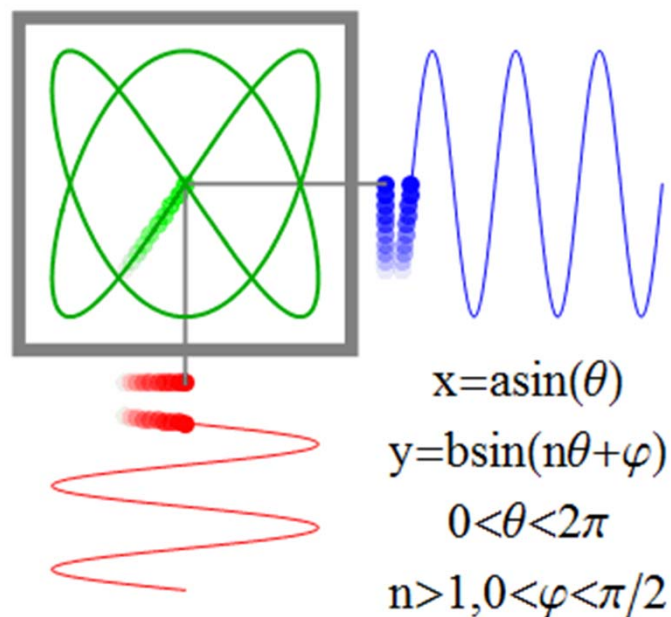
李萨如图1

任何一个直线上简谐振动, 匀角速度椭圆运动或匀速圆周运动都可分解为两个相互垂直的简谐振动.₂₃

2、两相互垂直, 不同频率简谐振动的合成

- (1). 若 ω_1 与 ω_2 不成比例关系, 图形不断变化, 曲线不封闭.
- (2). ω_1/ω_2 有整数比 (有理数) 的关系.

李萨如图形



李萨如图2

本章结束!